MEDIKIOSK CLINICAL PLATFORM DOCUMENTATION

TOOLS AND TECHNOLOGIES

Tools and Technologies Inventory

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Technical Inventory Table

Category	Tool / Technology	Version	Purpose	Where Used (File Paths)	Evidence / Lockfile Reference
Programming Languages	TypeScript	`^5.5.4` (backend) `~6.0.2` (frontend)	Primary strongly-typed language for backend and frontend codebase	`backend/package.json:L48`, `frontend/package.json:L42`	`package.json` dependencies
Programming Languages	JavaScript (Node.js ES Modules & CommonJS)	Node `22`	Server runtime, build scripts, demo automation scripts	`.node-version:L1`, `backend/Dockerfile:L2`, `railway.json:L9`	`.node-version`, `Dockerfile`
Programming Languages	Python	`3.9+`	Fine-tuning scripts and evaluation routines	`training/qlora_finetune.py`	`qlora_finetune.py`
Programming Languages	SQL (PostgreSQL Dialect)	PostgreSQL 15+	Relational database schema, vector extensions, and migration scripts	`backend/schema.sql:L1-L174`, `survival_migration.sql`	`schema.sql`, `survival_migration.sql`
Frontend Frameworks	React	`^19.2.8`	Component-based user interface framework for patient kiosk & doctor portal	`frontend/package.json:L20`	`frontend/package.json`
Frontend Frameworks	React DOM	`^19.2.8`	Virtual DOM renderer for React web application	`frontend/package.json:L21`	`frontend/package.json`
Routing	React Router DOM	`^7.18.3`	Client-side routing across kiosk screens and doctor dashboard	`frontend/package.json:L22`	`frontend/package.json`
Build Tools	Vite	`^8.2.2`	Next-generation frontend bundler and local development server	`frontend/package.json:L43`, `frontend/vite.config.ts`	`frontend/package.json`
Backend Framework	Express	`^5.2.1`	HTTP web server framework for REST endpoints and middleware	`backend/package.json:L23`, `backend/src/app.ts:L26`	`backend/package.json`
Backend Execution	`tsx`	`^4.23.13`	TypeScript execute and watch runtime for Node.js	`backend/package.json:L7`, `backend/package.json:L47`	`backend/package.json`

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Styling & Design System	Tailwind CSS	`^3.4.19`	Utility-first CSS framework for Ayurvedic theme styling	`frontend/package.json:L41`, `frontend/tailwind.config.ts`	`frontend/package.json`
Styling Utilities	`clsx` & `tailwind-merge`	`^2.1.1` / `^3.6.0`	Conditional class composition and Tailwind rule merging	`frontend/package.json:L17`, `frontend/package.json:L23`	`frontend/package.json`
Iconography	Lucide React	`^1.39.0`	Modern SVG icon set for UI components	`frontend/package.json:L19`	`frontend/package.json`
UI Primitives	Radix UI Slot	`^1.3.3`	Unstyled accessible UI primitive slots for component wrapping	`frontend/package.json:L14`	`frontend/package.json`
Database & Realtime	Supabase JS Client	`^2.114.0`	PostgreSQL ORM client, auth verification, and WebSocket Realtime subscriptions	`backend/package.json:L18`, `frontend/package.json:L15`, `backend/src/supabase.ts`	`package.json` lockfiles
Database Extensions	Supabase `pgvector`	`0.5.0+` (TODO: verify)	Vector similarity search for RAG embeddings in `rag_chunks` table	`backend/src/rag/retriever.ts:L39-L115`	`backend/src/rag/preseed.ts`
Client-Side Storage	`idb-keyval`	`^6.3.0`	Lightweight IndexedDB key-value wrapper for offline patient answer buffering	`frontend/package.json:L18`	`frontend/package.json`
Cloud AI SDK	Google Generative AI	`^0.24.1`	Client library for Gemini 3.6 Flash / 3.5 Flash LLM synthesis and embeddings	`backend/package.json:L17`, `backend/src/app.ts:L153`	`backend/package.json`
Speech & Vision API	Sarvam AI API	`v3` / `v1`	External REST API for Saaras v3 ASR, Bulbul v3 TTS, Doc AI digitise OCR, Mayura v1 Translation	`backend/src/orchestration/sarvamClient.ts:L1-L119`	`backend/src/orchestration/sarvamClient.ts`
Local LLM Runtime	Ollama	`latest` (TODO: verify)	Background service manager running `medgemma1.5:latest` for local inference	`backend/src/services/ollamaBootstrap.ts:L1-L56`	`backend/src/services/ollamaBootstrap.ts`
Client & Fallback OCR	Tesseract.js	`^7.0.0`	Pure JS / WebAssembly OCR engine for prescription text extraction	`backend/package.json:L29`, `frontend/package.json:L25`	`package.json` files

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Python ML Frameworks	PyTorch	`2.0+` (TODO: verify)	Deep learning framework for QLoRA fine-tuning	`training/qlora_finetune.py:L8`	`training/qlora_finetune.py`
Python ML Libraries	Hugging Face `transformers`, `peft`, `trl`, `datasets`, `bitsandbytes`	TODO: verify	Model loading, LoRA adapter injection, 4-bit quantization, SFT Trainer	`training/qlora_finetune.py:L9-L17`	`training/qlora_finetune.py`
Authentication & Security	`bcryptjs`	`^3.0.3`	Salted password hashing (10 rounds) for doctor accounts	`backend/package.json:L20`, `backend/src/app.ts:L5`	`backend/package.json`
Security Middleware	`cors` & `helmet` security headers	`^2.8.6`	Cross-Origin Resource Sharing control and HTTP header hardening	`backend/package.json:L21`, `backend/src/middleware/security.ts`	`backend/package.json`
Payload Validation	Zod	`^4.6.4`	TypeScript-first schema validation for API request bodies	`backend/package.json:L31`, `backend/src/schemas/index.ts`	`backend/package.json`
File Upload Handling	Multer & Form-Data	`^2.3.0` / `^4.0.6`	Middleware for handling `multipart/form-data` audio/document uploads	`backend/package.json:L24-L25`, `backend/src/app.ts:L157`	`backend/package.json`
E2E Browser Testing	Playwright	`^1.63.0` (frontend) `^1.62.1` (backend)	End-to-end user journey, voice recognition, and real-time verification testing	`frontend/package.json:L29`, `backend/package.json:L27`, `frontend/playwright.config.ts`	`package.json` files
Unit & Integration Testing	Vitest & Jest	`^4.1.11` (vitest) `^30.5.1` (jest)	Frontend unit runner (Vitest) and backend API route runner (Jest + Supertest)	`frontend/package.json:L44`, `backend/package.json:L42`	`package.json` files
Accessibility Testing	`@axe-core/playwright`	`^4.13.0`	Automated WCAG accessibility auditing in Playwright tests	`frontend/package.json:L28`	`frontend/package.json`
Linting & Quality	Oxlint	`^1.79.0`	High-performance Rust-based JavaScript/TypeScript linter	`frontend/package.json:L39`, `frontend/.oxlintrc.json`	`frontend/package.json`

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Containerization	Docker & Docker Compose	Docker API v2	Multi-stage Docker building Node 22 Alpine and Nginx Alpine containers	`Dockerfile` (backend), `Dockerfile` (frontend), `docker-compose.yml`	`docker-compose.yml`
Cloud Deployment	Railway & Vercel	Nixpacks / Vite	Cloud hosting configuration manifests for backend API and frontend static assets	`railway.json`, `vercel.json`, `frontend/vercel.json`	Configuration JSON files
CLI Automation	Concurrently	`^9.0.0`	Running backend and frontend dev servers in parallel during local demo	`package.json: L15`	Root `package.json`